

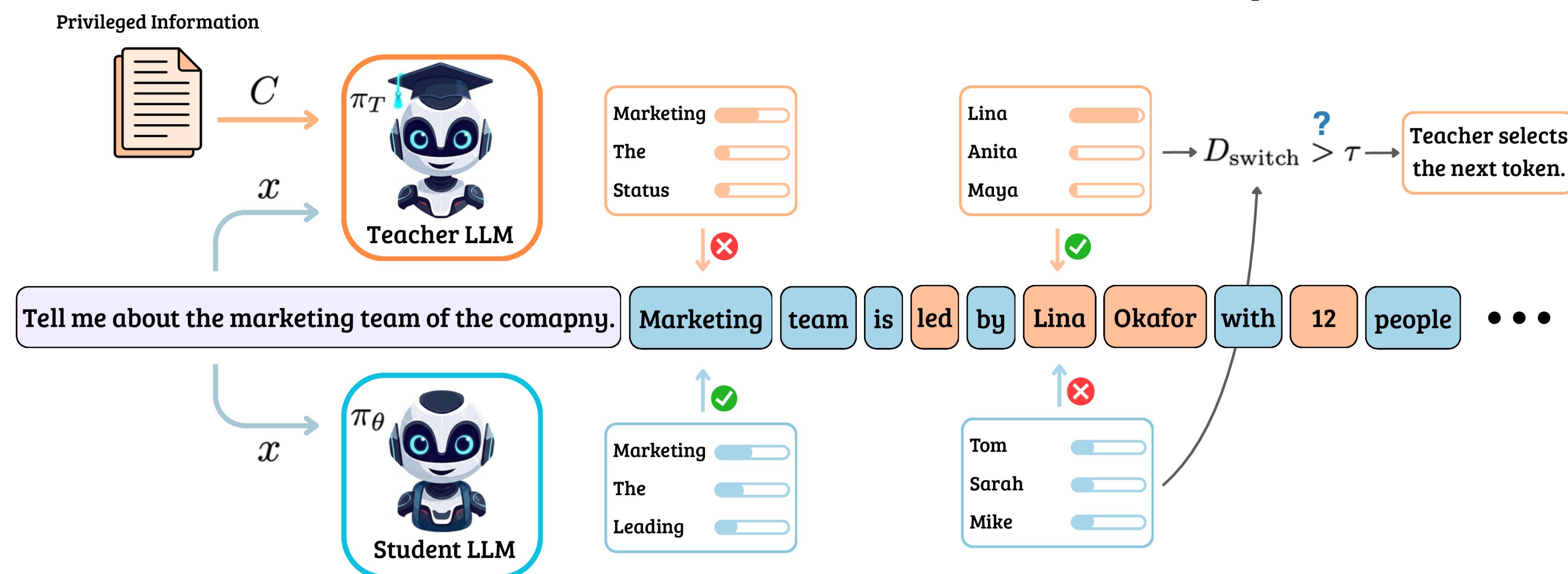
Speculative Self-Distillation enables Efficient Knowledge Internalization

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TL;DR

- 1 Loading context into the models
- 2 Generating the next token distribution from both LLMs
- 3 Deciding which distribution to sample from based on the divergence



To internalize a document, **self-distillation** trains a model to match itself with the document in context.

SSD lets the student write each rollout and hands over to the teacher **only at tokens where the two disagree**.

It matches **on-policy** accuracy with 45% fewer supervised tokens and **forgets less**.

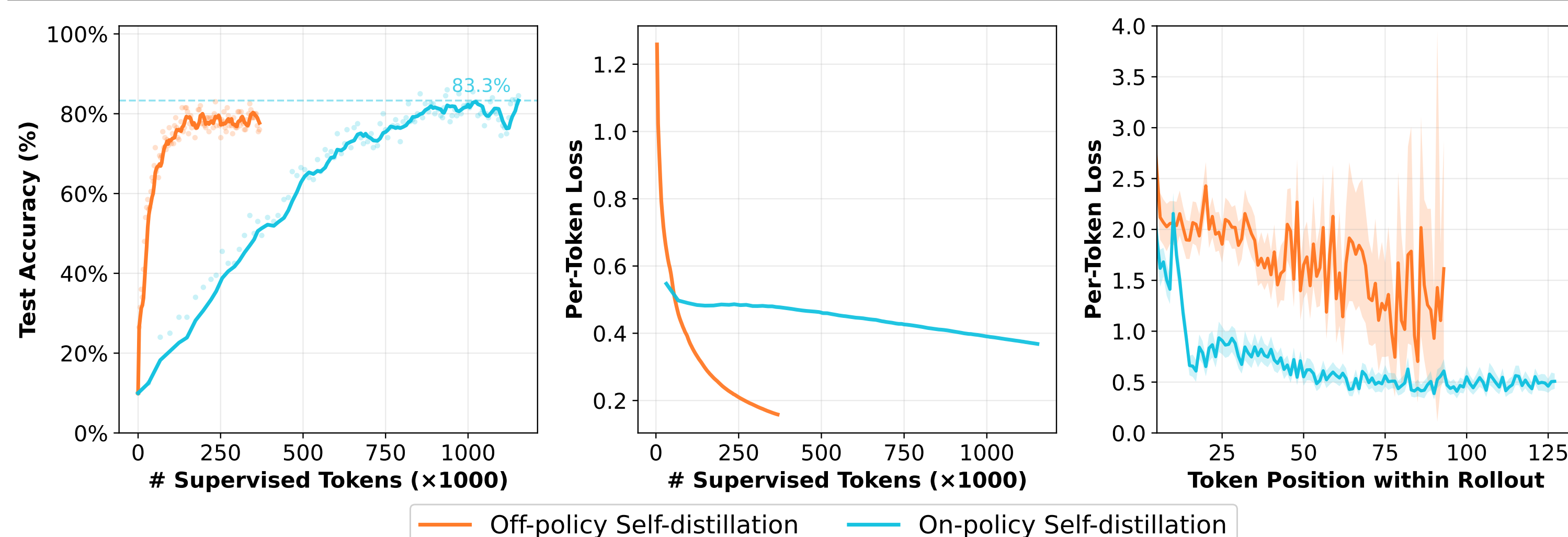


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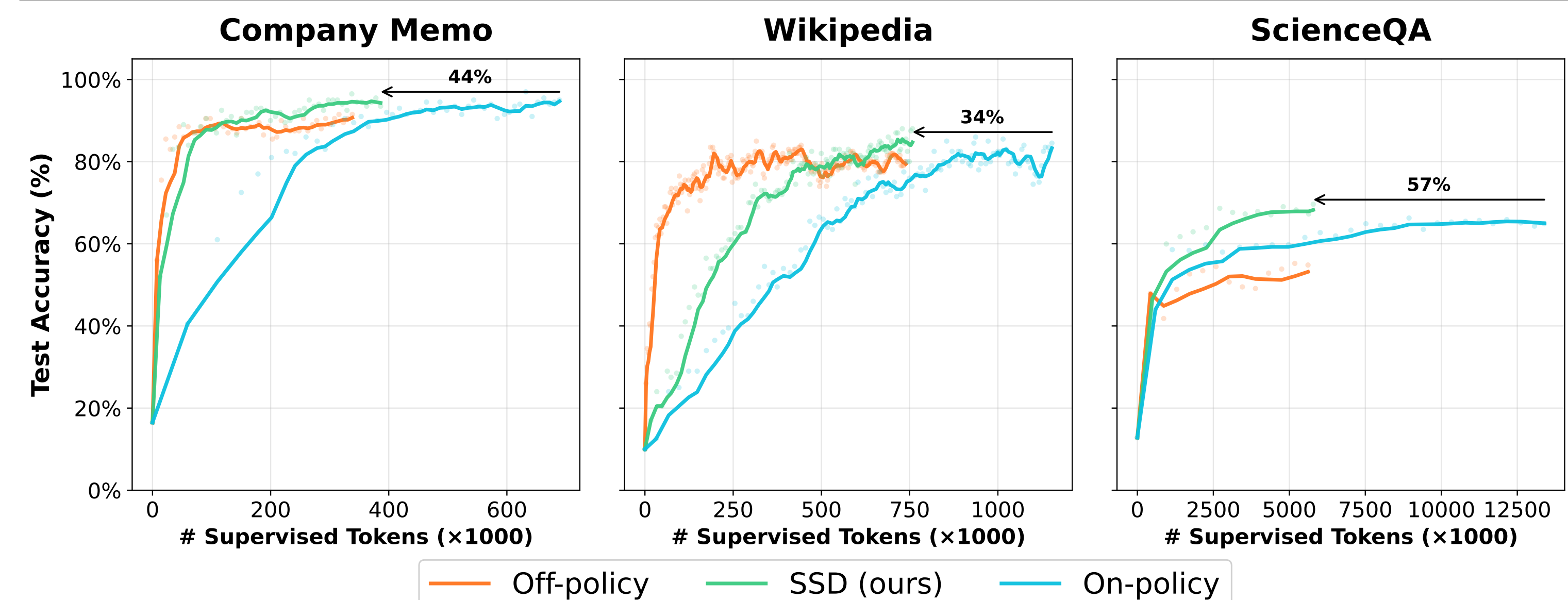
BLOG

On-policy rollouts waste most of their supervision



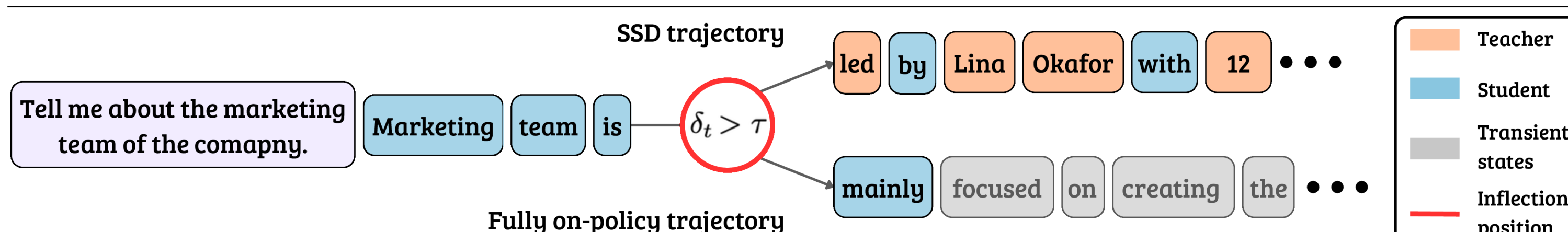
off-policy learns $\sim 5\times$ faster but plateaus lower. Along **on-policy** rollouts the per-token signal collapses after a few tokens.

SSD reaches on-policy accuracy with 34–57% fewer tokens



Qwen3-4B-Instruct as student and teacher on three closed-book QA tasks.

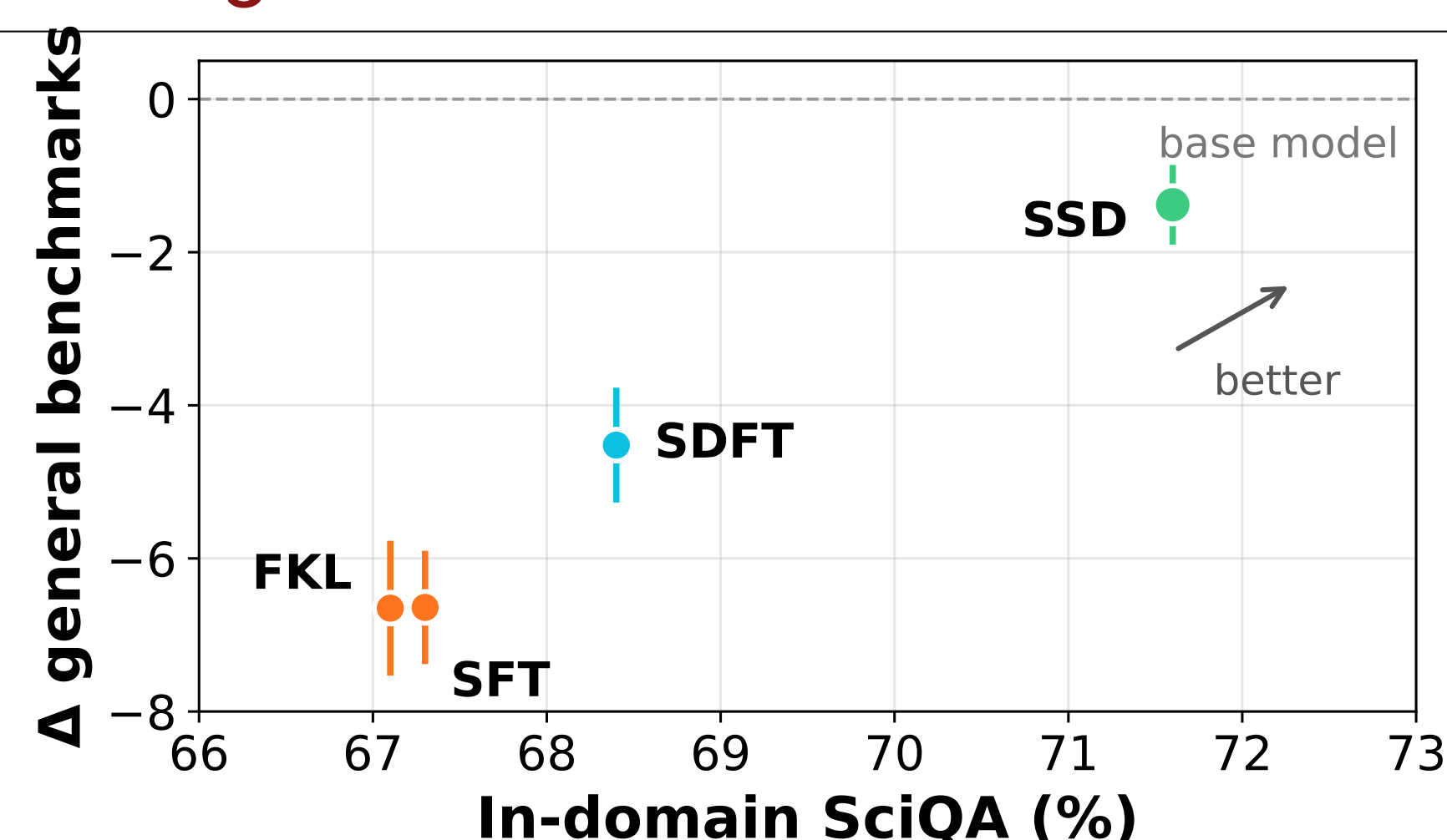
Inflection tokens: where the teacher matters



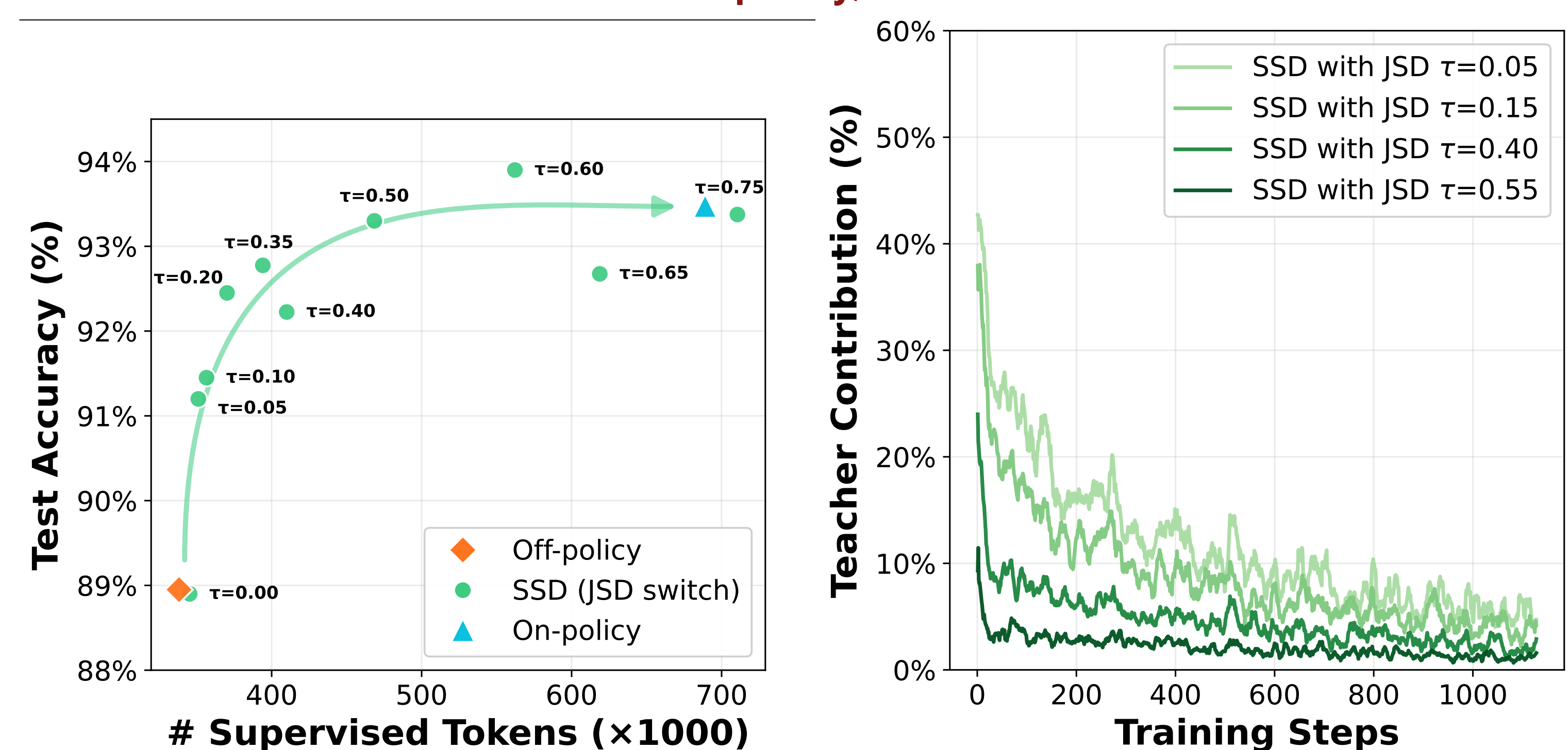
Where $\delta_t > \tau$ the teacher writes; **on-policy** drifts into transient states.

SSD forgets less

Qwen2.5-7B on SciKnowEval chemistry. **SSD** has the best in-domain accuracy and the smallest drop on six general benchmarks (-1.4 vs -4.5 for SFT).



One knob from off- to on-policy, and a built-in curriculum



$\tau \rightarrow 0$ is **off-policy**, $\tau \rightarrow \infty$ is **on-policy**. $\tau \in [0.2, 0.5]$ sits on the frontier.

The teacher's share of tokens falls during training with no schedule.